Real frozen-cache validation sequences: static COT can apply the same optical correction with opposite GHI outcome

+45 min | +60 min | +75 min | +90 min

Predicted AGRI false-colour

Predicted log1p(COT)

+105 min | +120 min | +135 min | +150 min

Predicted AGRI false-colour

Predicted log1p(COT)

frozen R predicted log1p(COT)

3.6
3.4
3.2
3.0
2.8
2.6

COT correction reduces error

Station: Zhujia
Target lead: +90 min
station COT at [8,8]: 3.33

Observed GHI: 344.3 W m^{-2}
AGRI-only: 223.0
AGRI + static COT: 255.7
COT correction: +32.8

Same type of static COT dimming can move one case toward the observed GHI and another away from it. The static map does not encode the missing cloud evolution.

COT correction increases error

Station: Sili
Target lead: +150 min
station COT at [8,8]: 3.56

Observed GHI: 364.7 W m^{-2}
AGRI-only: 291.5
AGRI + static COT: 245.8
COT correction: -45.7

Same type of static COT dimming can move one case toward the observed GHI and another away from it. The static map does not encode the missing cloud evolution.

All maps are actual predicted AGRI/COT from saved validation rows. AGRI display uses a per-sequence robust false-colour stretch; it does not alter model inputs. Cases are predeclared median-magnitude outcomes among high station-centre COT, top-decile C